Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6				**Indicative content** • removal of temporary hardness by boiling • hydrogencarbonate ions are not thermally stable and decompose easily on heating • this forms a layer of calcium carbonate (inside kettles) • calcium hydrogencarbonate → calcium carbonate + water + carbon dioxide • $Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$ • boiling does not remove permanent hardness • removal of permanent hardness by adding washing soda/sodium carbonate • this forms a (white) precipitate • sodium carbonate + calcium sulfate → calcium carbonate + sodium sulfate • $Na_2CO_3 + CaSO_4 \rightarrow CaCO_3 + Na_2SO_4$	6			6		3
				5-6 marks Detailed description of how hard water is softened using both methods; one correct equation *There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.* **3-4 marks** Good description of how water is softened using both methods *There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.* **1-2 marks** Brief description of how water is softened using one of the methods *There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.* **0 marks** *No attempt made or no response worthy of credit.*						
				Question 6 total	**6**	**0**	**0**	**6**	**0**	**3**